SPRMamba: A Mamba-Based Saliency Proportion Reconciliatory Network With Squeezed Windows for Remote Sensing Change Detection

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***Abstract*— Remote sensing change detection (RSCD) faces challenges in effectively identifying nonsalient change regions, such as subtle architectural modifications or changes that closely resemble the background. The primary difficulties stem from the weak feature representation of nonsalient changes, which results in insufficient model response, and the high similarity between background and change regions, leading to misdetection or omission. To address this issue, we propose SPRMamba, a Mamba-based saliency proportion reconciliatory network with squeezed windows for RSCD. It introduces state-space models (SSMs) with windowing operations and cross-window interaction mechanisms to improve the response to weak signals. To dynamically balance the representation of salient and nonsalient features, we design the saliency proportion reconciler (SPR) to optimize the discrimination between background and change regions. In addition, we introduce a sparse saliency loss function, which imposes sparsity constraints on salient regions to enhance the feature representation of nonsalient change regions. The experimental results show that SPRMamba significantly outperforms existing methods on several public datasets. Our code will be available at https://github.com/boomstarzzn/SPRMamba**

***Index Terms*— Nonsalient changes, remote sensing change detection (RSCD), saliency proportion reconciliation, state-space model (SSM).**

I. Introduction

CHANGE detection (CD) aims to identify pixel-level changes in objects or phenomena by comparing imagery data of the same area across different time periods. In the context of remote sensing (RS), this technique plays a pivotal role in various domains such as urban expansion monitoring, land use analysis, and environmental protection [1], [2], [3], [4].

In the early stages of CD research, traditional methods (e.g., algebraic difference methods [5], transformation-based detection methods [6], and classification-based methods [7]) primarily relied on handcrafted low-level features such as spectral differences and texture characteristics. While these methods performed well in simple scenarios and salient CD, their robustness significantly declined when facing challenges such as illumination variation, seasonal effects, and complex background interference [8].

In recent years, deep learning [9], [10], [11], [12], [13], [14], [15], [16] (e.g., convolutional neural networks (CNNs) [17], [18], [19], [20], [21], [22] and Transformers [23], [24], [25], [26], [27]) has substantially improved CD performance through data-driven feature extraction. Due to their robust local feature extraction capabilities, CNNs are highly effective at capturing building edges and texture information. In contrast, Transformers leverage global self-attention mechanisms to model long-range dependencies, making them particularly advantageous for large salient CD. Moreover, the emergence of state-space models (SSMs) [28], [29], [30], [31], [32], [33], [34], [35], especially the Mamba architecture, introduces a novel approach to CD by employing raster scanning mechanisms to achieve low-complexity global information modeling.

Despite their advancements, these methods predominantly focus on detecting salient changes (e.g., large-scale building construction, demolition, or significant disaster-related changes) and still face critical challenges when addressing nonsalient changes (e.g., minor building modifications or changes similar to the background), as shown in Fig. 1. The main issues include the following.

1) *Insufficient Feature Representation:* Nonsalient change regions typically exhibit weak features (e.g., slight variations in pixel values or color gradients), which tend to be diluted or overlooked during multilayer propagation in deep networks.
2) *Background Feature Similarity:* Nonsalient change regions often share high spectral and texture similarity with the background, making it difficult for existing methods to accurately differentiate these regions, thereby resulting in false positives (FPs) or missed detections.

Therefore, effectively detecting both salient and nonsalient changes while ensuring balanced feature representation

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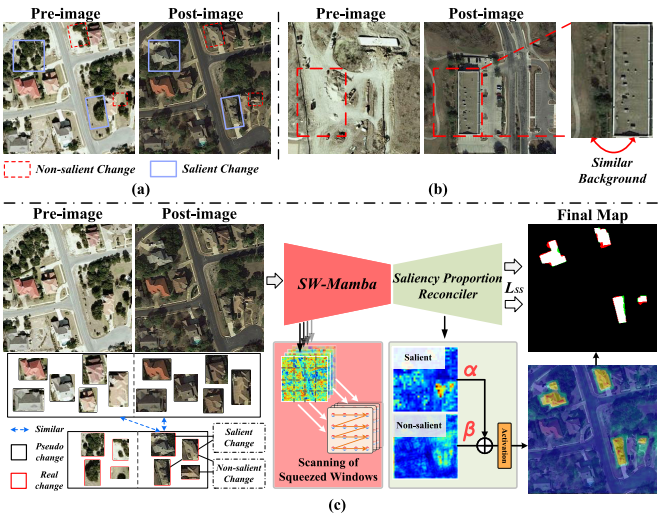


Fig. 1. (a) Challenge of detecting subtle change regions in CD tasks. (b) Issue of background similarity, where true change regions with colors similar to the background are prone to omission during detection. In (a) and (b), the red dashed boxes indicate nonsalient changes, while the blue solid boxes represent salient changes. (c) Schematic of SPRMamba. It employs the proposed SW-Mamba to extract comprehensive change information and utilizes SPR to dynamically adjust the proportion of salient and nonsalient features.

is recognized as a complex task. Traditional architectures like CNNs often lack sufficient global contextual modeling, while Transformers, despite excelling at long-range dependencies, suffer from quadratic computational complexity [31]. Moreover, their softmax-based attention mechanism can inadvertently suppress weaker, nonsalient change features due to exponential normalization, leading to their decay during interlayer propagation [36]. In contrast, Mamba [37] offers compelling advantages through its linear complexity and selective state-space modeling, as shown in Fig. 2. Its selective scanning mechanism preserves weak signals via input-dependent gating and linear feature scaling, thereby mitigating the feature decay issue often observed in attention mechanisms and establishing it as a promising foundation for balanced feature extraction in RSCD. However, standard Mamba's sequence flattening operation can disrupt vital spatial neighborhood relationships within RS imagery, hindering the preservation of fine-grained local features crucial for detecting subtle, nonsalient changes. While approaches like multidirectional scanning [e.g., in Vision Mamba (ViM)] attempt to alleviate this, the fundamental challenge of reconciling global sequence modeling with local spatial integrity, particularly for capturing the nuanced characteristics of nonsalient changes within defined local spatial contexts, remains an area requiring further improvement.

To address the aforementioned issues, we designed SPRMamba, which optimizes the balanced representation of salient and nonsalient features while ensuring comprehensive image feature acquisition. First, we developed the SW-Mamba backbone, composed of window-constrained SSM (SW-SSM). This backbone performs modeling of both window-level details and cross-window contexts within spatially coherent regions, thereby preserving local geometric structures while achieving global contextual awareness. Subsequently,



Fig. 2. Comparison of the self-attention mechanism and selective scanning mechanism. (a) Self-attention mechanism. (b) Cross-scan mechanism.

we introduced the saliency proportion reconciler (SPR), which dynamically adjusts the feature weights of salient and nonsalient regions. This ensures that the model does not overlook nonsalient changes while focusing on salient ones. The synergy between SW-Mamba and SPR adaptively integrates multiscale features, thereby enhancing the comprehensiveness and accuracy of detection.

The main contributions of this article are as follows.

1) We design the SW-Mamba backbone network, which incorporates window operation and cross-window interaction mechanisms, to solve the perceptual challenge in nonsalient CD. It significantly improves the ability of the model to detect global and local features and effectively reduces the detection miss caused by feature fragmentation or background interference.
2) We design SPR to balance the feature representation of salient and nonsalient regions by dynamically adjusting the feature weights of the two types of regions. It significantly enhances the model's ability to detect changes in nonsalient regions so that it can still accurately capture subtle changes in complex scenes.
3) We introduce a sparse saliency loss function to strengthen the representation of nonsalient features and suppress the overresponse in salient regions from the loss level. It effectively balances the feature response of the real change region through the sparsity constraint, especially in scenes dominated by nonsalient changes.

II. Related Work

A. CNN-Based Methods

CNNs have been widely applied to CD tasks due to their hierarchical feature extraction capabilities [38], [39], [40], [41], [42], [43], [44], [45], [46], [47], [48], [49], [50]. Seminal models such as FC-EF and FC-Siam-Di [19] leverage UNet-like architectures [51], [52] to perform multiscale feature extraction and generate change maps. To enhance the perceptual accuracy of change regions, attention mechanisms have also been incorporated into CD methodologies. For instance, SNUNet [53] integrates dense connections with channel attention for salient feature enhancement; DMINet [54] employs joint temporal attention to improve the localization of prominent changes.

Although CNNs excel in handling local details, they face challenges in capturing cross-region semantic relationships in CD tasks due to their inherent local convolutional nature [55], [56], [57]. As shown in Fig. 1, when the nonsalient change

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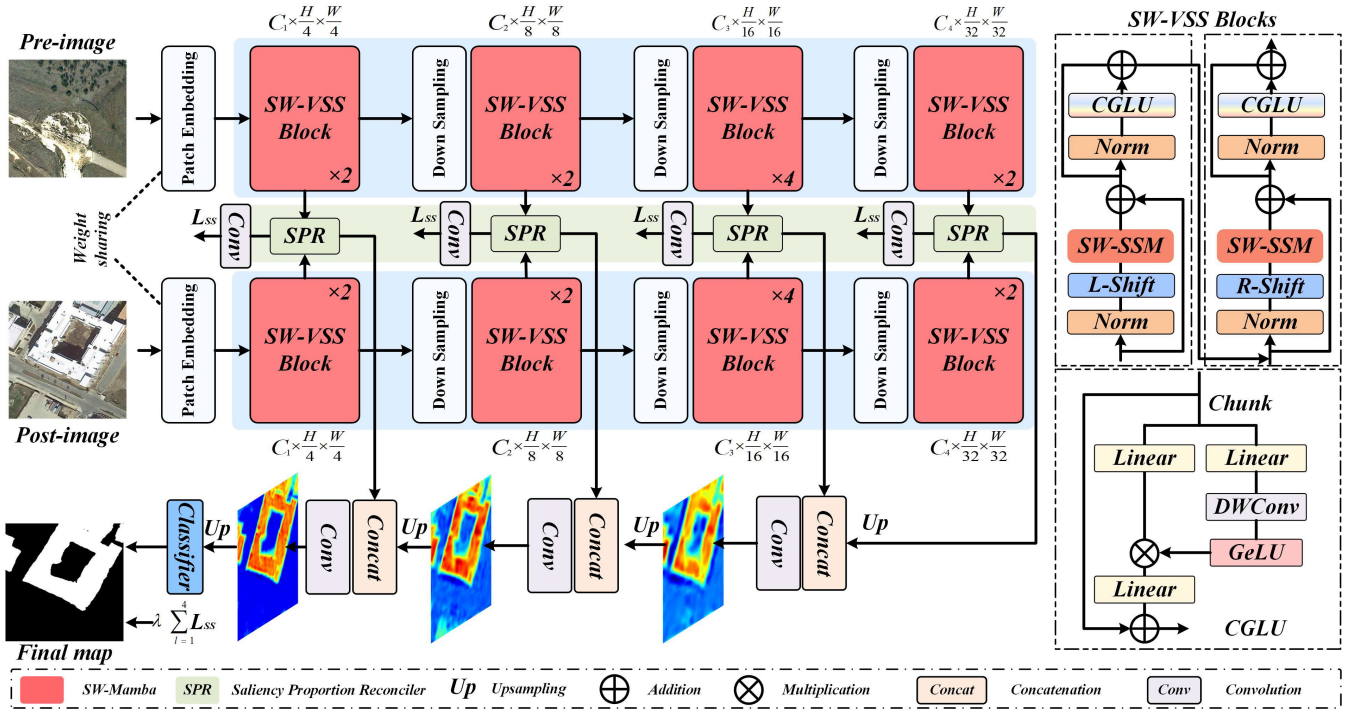


Fig. 3. Overall architecture of the proposed SPRMamba. It primarily includes: 1) SW-Mamba backbone and 2) SPR.

areas share high similarity with the background features, misjudgments or omissions are more likely to occur. Additionally, CNN's feature extraction and response mechanism tends to prioritize salient change areas, causing weak features from nonsalient change regions to gradually diminish during hierarchical propagation. This makes it difficult for CNNs to achieve a balanced representation between salient and nonsalient areas, highlighting the need for novel architectures that balance local accuracy with long-range dependency perception.

B. Transformer-Based Methods

Transformers, with their self-attention mechanisms, effectively model long-range dependencies in images, making them an increasingly prevalent tool in CD tasks [58], [59], [60], [61], [62], [63], [64], [65], [66]. For example, BIT [25] integrates semantic labels with Transformer modules to enhance spatiotemporal modeling capabilities. ChangeFormer [24] utilizes pure Transformer blocks as the backbone for end-to-end CD. MDIPNet [26] improves semantic consistency through lightweight parallel fusion and cross-space perception.

Although Transformers address the global modeling limitations of CNNs, they still face challenges in handling nonsalient changes. Their self-attention mechanism relies on an attention matrix for feature fusion, where salient change regions often attract more attention due to their stronger feature activation. In contrast, nonsalient change regions, with weaker feature signals or those overshadowed by salient features, can lead to reduced sensitivity to subtle changes. This characteristic makes it difficult for Transformers to achieve a balanced modeling of salient and nonsalient changes. Moreover, their high computational complexity poses additional challenges for processing large-scale RS imagery.

C. SSM-Based Methods

SSMs are emerging as a significant research direction beyond Transformers due to their advantageous linear computational complexity in processing sequential data [67]. Early works, such as the S4 model [28], demonstrated the potential of SSMs in long-sequence modeling through structured state matrices and efficient convolutional representations. Building upon this foundation, Mamba [37] introduces a core selective mechanism, enabling the model to dynamically adjust its internal parameters based on the input. This input-dependent characteristic allows Mamba to selectively focus on critical information within the sequence while filtering out irrelevant content, thereby enhancing its expressive power and contextual adaptability while maintaining global information capture capabilities. Its efficient hardware-aware design also ensures high performance in processing long sequences. To adapt Mamba for 2-D visual tasks, ViM [29] flattens image patches into sequences and employs multidirectional scanning strategies to capture spatial context. Its optimized selective scanning module (OSSM) has demonstrated strong competitiveness in various vision tasks.

Mamba offers a novel perspective for RSCD, demonstrating particular proficiency in processing large-scale, high-resolution imagery and identifying nonsalient changes. Unlike CNNs that rely on local convolutions, Mamba emphasizes the contextual understanding and evolving trends of changes through state transmission along scan paths. Furthermore, in contrast to the global attention mechanisms of Transformers, Mamba employs structured state recursion and selective scanning to efficiently aggregate information and selectively preserve contextual dependencies.

Recently, Mamba has been explored for application in CD [68], [69]. For instance, RS-Mamba [30] leverages the

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OSSM to efficiently model global context, capturing more comprehensive spatial features. ChangeMamba [31] utilizes ViM as its encoder, further validating the potential of SSMs in various CD tasks. RSCaMa [32] proposed a spatial difference-aware and temporal traversal SSM, which effectively enhances the model's ability to capture spatial variations and comprehend temporal interactions.

Despite its excellent performance in global modeling, when directly applied to RSCD, Mamba's sequential scanning mechanism still falls short in capturing the inherent 2-D isotropic spatial structure of images, which limits its sensitivity to feature expression in regions with nonsalient changes. This indicates that relying on the global modeling capability of standard Mamba alone is not sufficient to fully solve the problem of balanced representation of salient and nonsalient change regions in RSCD.

III. Methodology

In this section, we first describe the network architecture of SPRMamba. Next, we provide the detailed design of the SW-Mamba based on SSM. Following that, we elaborate on the proposed SPR, which is primarily implemented using saliency bias control factors, the saliency change memory mechanism, and the spatial filter. Finally, we introduce the implementation of the sparse saliency loss, which aims to enhance the representation of nonsalient changes.

A. Overview

As shown in Fig. 3, the overall pipeline of the proposed SPRMamba is designed based on a weight-sharing encoder–decoder framework. Given the bi-temporal RS images T_1, $T_2 \in \mathbb{R}^{3\times H\times W}$, the hierarchical SW-Mamba backbone is used to extract multiscale feature maps with channel dimensions of 64, 128, 256, and 512. The extracted bi-temporal feature maps are then fused using the SPR, which introduces the saliency memory mechanism to dynamically generate saliency and nonsaliency weight maps, thereby reconciling the contributions of salient and nonsalient changes. Subsequently, the fused multiscale feature maps undergo progressive upsampling and feature aggregation operations. Finally, the classification head generates the pixel-level high-resolution change map (Final map).

B. SW-Mamba Backbone

The core of SW-Mamba lies in the combination of windowing operations and an SSM-based window interaction mechanism, as shown in Fig. 4. This design integrates the long-range dependency awareness of raster scanning with local fine-grained modeling within windows, thereby achieving balanced modeling of salient and nonsalient changes during the feature extraction stage. Specifically, in Fig. 5, the input feature is denoted as $X \in \mathbb{R}^{H\times W\times C}$. First, the input X is divided into windows of size $w \times w$, where each window is represented as $x_i \in \mathbb{R}^{w\times w\times C}$, which can be expressed as

$$X_{\text{windows}} = \{x_1, x_2, \ldots, x_n\},\ x_i \in \mathbb{R}^{w\times w\times C} \tag{1}$$

where n denotes the number of windows along the horizontal or vertical direction in the image, with $n = (H/w) = (W/w)$, meaning that the image is divided into $(H/w) \times (W/w)$ windows. $n^2 \times w \times w \times C$ represents the shape of all the window features after unfolding, n^2 denotes the total number of windows, w is the side length of each window, and C is the number of channels.

To overcome the limitation of traditional windowing operations that restrict global information interaction between windows, we introduce the SSM mechanism. The SSM enables long-range information flow between windows, allowing the model to establish global associations between windows while maintaining refined local features. Specifically, for each window $x_i \in \mathbb{R}^{w\times w\times C}$, we apply a pooling operation: $\text{Pool}(x_i) = \widetilde{x_i} = (\text{Maxpool}(x_i) + \text{Avgpool}(x_i)/2)$, the combination of max pooling and average pooling captures the saliency statistics within the window, denoted as $\widetilde{x_i}$. This represents the statistical information for each window after compression.

To facilitate interactions between windows, we model all $\widetilde{x}$ using SSM to obtain $x_{\text{global}} \in \mathbb{R}^{1\times 1\times C}$, which captures long-range dependencies. At the same time, the window set X_{windows} is processed through SSM to obtain $x_{\text{local}} \in \mathbb{R}^{w\times w\times C}$, which captures the semantic detail correlations within each window. The process is as follows:

$$x_{\text{global}} = \text{Re}\big(\text{SSM}\big(\widetilde{x}_1, \widetilde{x}_2, \ldots, \widetilde{x}_n\big)\big) \tag{2}$$

$$x_{\text{local}} = \text{SSM}(X_{\text{windows}}) \tag{3}$$

$$x_{\text{fused}} = x_{\text{local}} \cdot \sigma(x_{\text{global}}) \tag{4}$$

where $\text{Re}(\cdot)$ represents reshaping, $\sigma(\cdot)$ represents the sigmoid activation function, and the fused feature $x_{\text{fused}} \in \mathbb{R}^{w\times w\times C}$ is recombined into features, with the original spatial resolution of the feature map restored through window reconstruction.

Additionally, SW-Mamba further alleviates the window boundary effect by dynamically adjusting the window shifting direction at different levels, as shown in Fig. 5. Specifically, four directions (top left, top right, bottom left, and bottom right) are alternately applied to the odd and even layers of the squeezed-window visual state space (SW-VSS). This hierarchical dynamic window shifting mechanism effectively breaks the directional bias through multipath feature aggregation. The diversity of scanning paths ensures unbiased coverage of omnidirectional changes, reducing the interference of window boundaries on both salient and nonsalient regions.

SW-Mamba achieves an effective balance between salient and nonsalient feature representation through the deep integration of local windowing and interwindow interactions. This establishes a robust foundation of high-quality features for the subsequent SPR.

C. Saliency Proportion Reconciler

Specifically, we first compute the differential features D from the bi-temporal features $F_1, F_2 \in \mathbb{R}^{C\times H\times W}$, $D = |F_1 - F_2| \in \mathbb{R}^{C\times H\times W}$. Then, D is reconstructed for 1-D processing, and its shape becomes $D_{\text{re}} \in \mathbb{R}^{C\times N}$, $N = H \times W$. The reconstructed features are subsequently refined through layer normalization and an embedding module to adjust the feature

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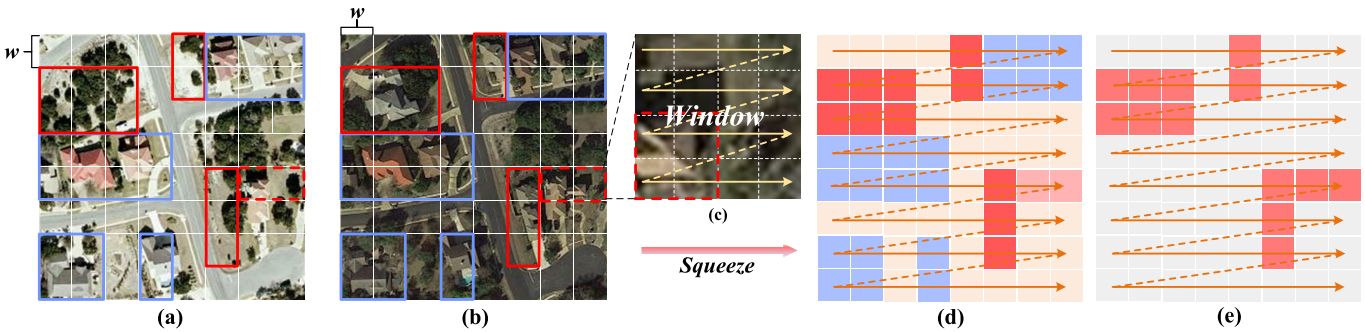


Fig. 4. Description of the SW-Mamba strategy. (a) and (b) Windowed bi-temporal inputs, with w denoting the window size, (c) local scan within a single window, (d) global scan across windows after squeezing, and (e) integrated features after scanning. In (a) and (b), the red solid box indicates salient changes, the red dashed box indicates nonsalient changes, and the blue solid box indicates pseudo changes.

TABLE I

EXPERIMENTAL RESULTS OF SPRMAMBA AND OTHER MODELS ON EACH DATASET. RED INDICATES THE HIGHEST VALUE FOR THE SELECTED METRICS, WHILE BLUE INDICATES THE SECOND-HIGHEST VALUE

Model	Publications	LEVIR-CD F1 / Pre / Rec / OA / IoU (↑)	WHU-CD F1 / Pre / Rec / OA / IoU (↑)	GZ-CD F1 / Pre / Rec / OA / IoU (↑)	Params(M)↓	Flops(G)↓
BIT [25]	TGRS·2022	89.31 / 89.24 / 89.37 / 98.92 / 80.68	87.47 / 88.71 / 86.27 / 98.81 / 77.73	80.23 / 82.40 / 78.18 / 96.31 / 66.99	3.55	10.60
ChangeFormer [24]	IGARSS·2022	90.40 / 92.05 / 88.80 / 99.04 / 82.48	86.88 / 88.50 / 85.33 / 98.76 / 76.81	73.66 / 84.59 / 65.23 / 95.53 / 58.30	41.03	202.87
FTN [70]	ACCV·2022	91.01 / 92.71 / 89.37 / 99.06 / 83.51	92.16 / 93.09 / 91.24 / 99.37 / 85.45	85.58 / 86.99 / 84.21 / 97.92 / 74.79	168.53	45.00
DMINet [54]	TGRS·2023	90.71 / 92.52 / 89.95 / 99.07 / 82.99	91.49 / 92.65 / 90.35 / 99.19 / 84.31	81.98 / 87.92 / 76.79 / 96.77 / 69.46	6.24	14.55
SNUNet [53]	GRSL·2021	88.16 / 89.18 / 87.17 / 98.82 / 78.83	88.34 / 91.34 / 85.53 / 98.91 / 79.11	83.64 / 87.55 / 80.05 / 97.00 / 71.87	12.03	54.88
AERNet [71]	TGRS·2023	90.78 / 89.97 / 91.59 / 99.07 / 83.11	92.18 / 92.47 / 91.89 / 99.25 / 85.49	84.42 / 88.06 / 81.07 / 97.13 / 73.03	25.36	12.82
RS-Mamba [30]	TGRS·2024	90.90 / 92.68 / 89.19 / 99.09 / 83.33	91.45 / 91.34 / 91.57 / 99.17 / 84.26	83.86 / 86.63 / 81.44 / 96.99 / 72.21	27.90	15.71
ChangeMamba [31]	TGRS·2024	90.16 / 91.59 / 88.78 / 99.01 / 82.09	93.33 / 94.76 / 91.94 / 99.52 / 87.49	84.26 / 89.04 / 79.98 / 97.08 / 72.81	17.13	45.74
CDMamba [34]	TGRS·2025	90.75 / 91.43 / 90.08 / 99.06 / 83.07	93.76 / 94.21 / 92.01 / 99.51 / 88.26	85.19 / 89.53 / 81.25 / 96.48 / 74.19	11.90	29.64
Ours(SPRMamba)	—	91.95 / 93.11 / 90.82 / 99.19 / 85.09	94.85 / 95.56 / 94.16 / 99.50 / 90.21	87.70 / 89.56 / 85.92 / 97.67 / 78.10	23.41	11.59

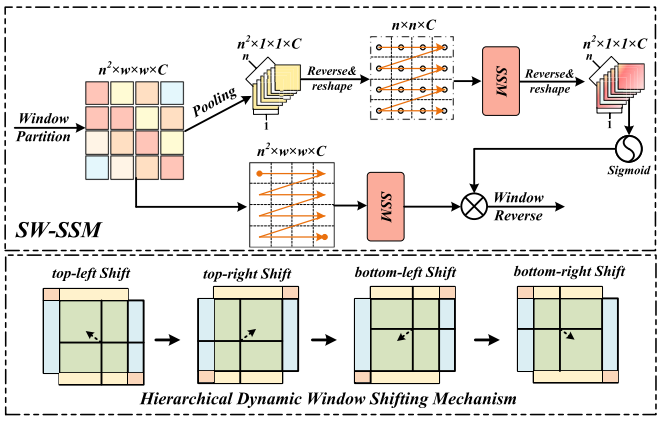


Fig. 5. Detailed description of SW-SSM and Hierarchical dynamic window shifting mechanism. Input features are partitioned into windows. Interwindow global context is captured by applying SSM to pooled window representations. Concurrently, intrawindow local details are extracted by applying SSM to the original windows. These global and local features are fused to the original resolution. To mitigate boundary effects, window shifting is applied in four alternating directions across different layers of the SW-VSS.

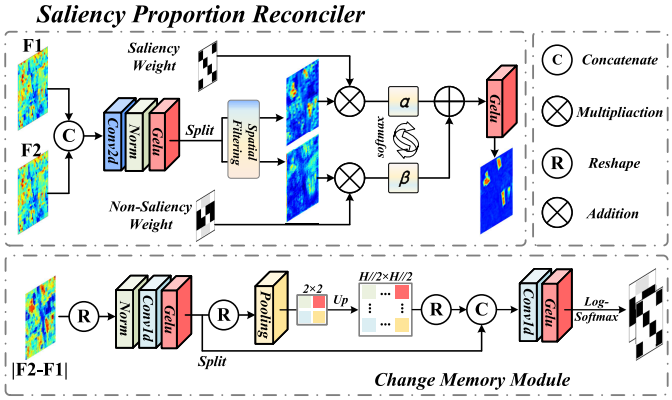


Fig. 6. Detailed description of SPR. The SPR's process goes through two main pathways. Change memory module derives saliency and nonsaliency weights from the feature difference. The top pathway processes bi-temporal features, applies these weights via spatial filtering and multiplication, and then dynamically fuses the weighted features using learnable parameters to produce the reconciled output.

representation. These processed features are then passed into the saliency memory design [33] for further operations, as shown in Fig. 6

$$D_{\text{embed}} = \mathcal{G}\big(\text{Conv1d}(\text{Norm}(D_{\text{re}}))\big) \tag{5}$$

$$D_1, D_2 = \text{Split}(D_{\text{embed}}, \text{dim} = C) \tag{6}$$

$$D_2' = \text{Pool}(D_2, \text{Size} = (2, 2)) \tag{7}$$

$$W_{\text{mix}} = \text{LogSoftmax}\big(\mathcal{G}\big(\text{Conv1d}\big(\oplus\big(D_1, \text{Up}(D_2')\big)\big)\big)\big) \tag{8}$$

where $\mathcal{G}(\cdot)$ represents the GeLU function; $\oplus(\cdot)$ represents the concatenation operation; and $W_{\text{mix}} \in \mathbb{R}^{2\times H\times W}$ is the comprehensive weight map of the differential features, used to generate the salient weight map $W_{\text{sal}} = [0, :, :] \in \mathbb{R}^{1\times H\times W}$ and the nonsalient weight map $W_{\text{nonsal}} = [1, :, :] \in \mathbb{R}^{1\times H\times W}$. On the other hand, the inputs are concatenated and passed through a convolution to obtain salient and nonsalient features

$$F_{\text{sal}}, F_{\text{nonsal}} = \text{Split}\big(\text{Conv2d}(\oplus(F_1, F_2)), \text{dim} = C\big) \tag{9}$$

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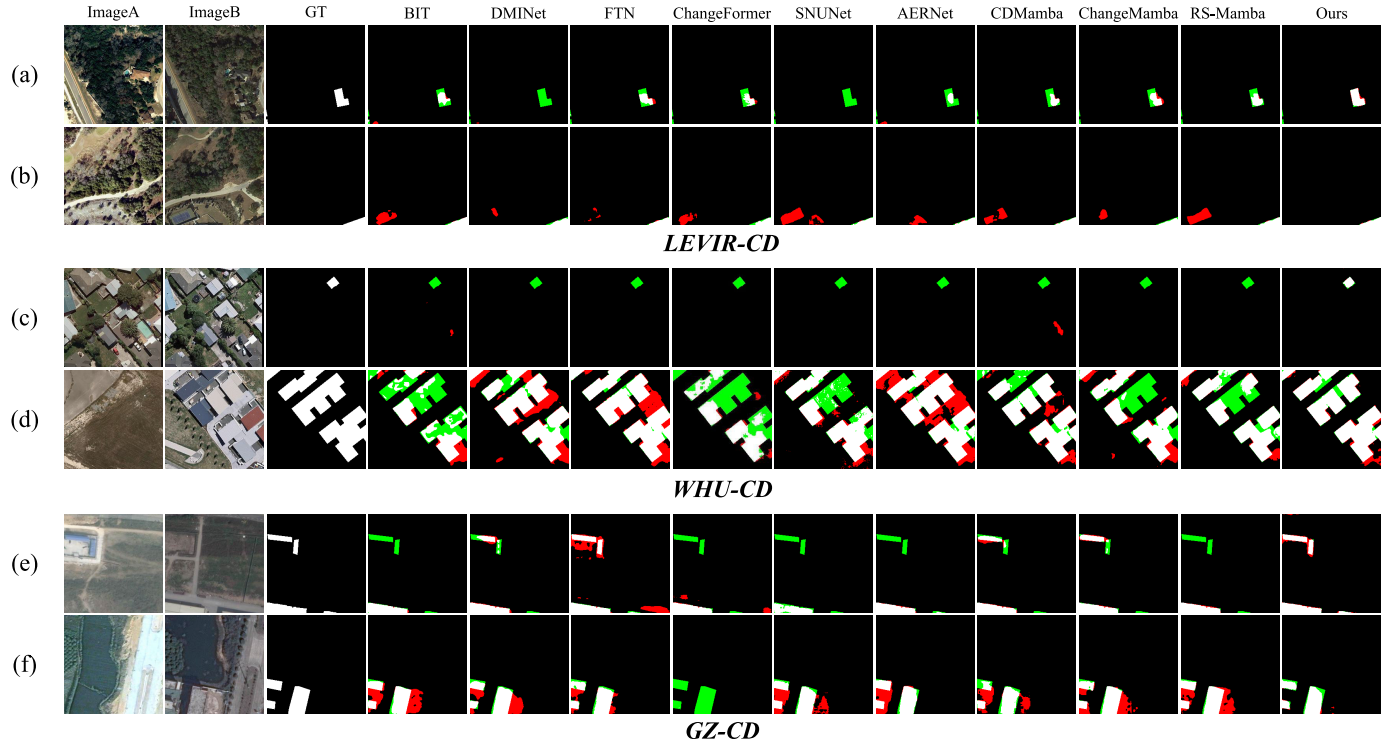


Fig. 7. Visualization results of subtle change and background-similar change areas selected from different datasets. (a) and (b) Results of different methods tested on LEVIR-CD, (c) and (d) results tested on WHU-CD, and (e) and (f) results tested on GZ-CD, where white represents the TP, black represents the TN, green represents the FN, and red represents the FP.

$$\widetilde{F}_{\text{sal}},\ \widetilde{F}_{\text{nonsal}} = SF\left(F_{\text{sal}}\right)\cdot W_{\text{sal}},\ SF\left(F_{\text{nonsal}}\right)\cdot W_{\text{nonsal}} \tag{10}$$

where SF(·) represents the spatial filter [72], designed to refine feature representation in the spatial dimension. The weights obtained from the differential features are applied to the corresponding fusions. Subsequently, dynamic fusion is performed based on the saliency bias control factors α and β, resulting in the output $F_{\text{out}} \in \mathbb{R}^{C\times H\times W}$

$$\alpha,\beta = \text{Softmax}\left([\alpha_{\text{param}},\beta_{\text{param}}]\right) \tag{11}$$

$$F_{\text{out}} = \mathcal{G}\left(\alpha\cdot\widetilde{F}_{\text{sal}} + \beta\cdot\widetilde{F}_{\text{nonsal}}\right). \tag{12}$$

SPR achieves effective modeling of both salient and nonsalient regions through a saliency memory design. Specifically, salient regions are highlighted with high weights to emphasize change features, while nonsalient regions are preserved using enhancement strategies to retain weak feature information. This enables a balanced representation of salient and nonsalient regions.

D. Loss Function

To further optimize the detection performance of salient and nonsalient change regions, we design a joint loss function. The loss function integrates the precise localization of salient regions, the preservation of details in nonsalient regions, and the sparsity constraint on the overall change regions, ensuring the model's generalization capability in complex scenarios. Specifically, the classic binary cross-entropy (BCE) loss function is first employed to measure the fundamental accuracy of the CD task

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N}\sum_{n=1}^{N}(y_n\log x_n + (1-y_n)\log(1-x_n)) \tag{13}$$

where N represents the total number of vectors and x_n and y_n represent the predicted value and label value of pixel n, respectively. On the other hand, the sparse saliency loss $\mathcal{L}_{\text{SS}}$ is designed to work in conjunction with the SPR to enhance the balanced constraint on salient and nonsalient change regions. Specifically

$$\mathcal{L}_{\text{s}} = \frac{1}{N}\sum_{i=1}^{N} W_s\cdot\|\psi(F_l)-Y\|_1 \tag{14}$$

$$\mathcal{L}_{\text{n}} = \frac{1}{N}\sum_{i=1}^{N} W_n\cdot\|\psi(F_l)-Y\|_2^2 \tag{15}$$

$$\mathcal{L}_{\text{SS}} = \mu\cdot\mathcal{L}_{\text{s}} + \theta\cdot\mathcal{L}_{\text{n}} \tag{16}$$

where W_s and W_n are the salient and nonsalient weight maps generated by the SPR, respectively, and $\psi(F_l)$ represents the features obtained through upsampling and convolutional dimensionality reduction. $\|\cdot\|_1$ denotes the $\mathcal{L}_1$-loss, used for sparse feature constraints, while $\|\cdot\|_2$ represents the $\mathcal{L}_2$-loss, used to ensure the smoothness of weak features. μ and θ control the constraint strengths for salient and nonsalient regions, respectively. The overall loss function is formulated as follows:

$$\mathcal{L}_{\text{T}} = \lambda_1\mathcal{L}_{\text{BCE}} + \lambda_2\sum_{l=1}^{4}\mathcal{L}_{\text{SS}} \tag{17}$$

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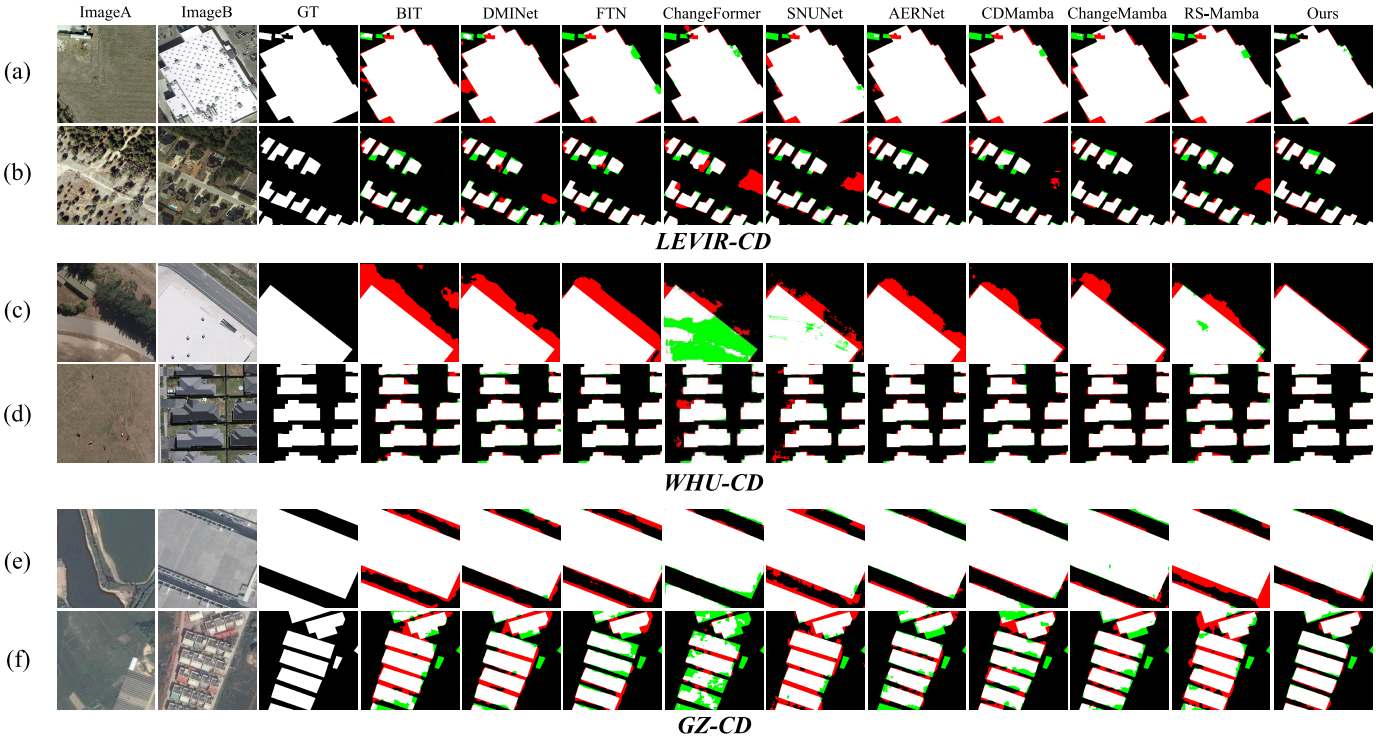


Fig. 8. Visualization results of large-scale change and dense region change areas selected from different datasets. (a) and (b) Results of different methods tested on LEVIR-CD, (c) and (d) results tested on WHU-CD, and (e) and (f) results tested on GZ-CD, where white represents the TP, black represents the TN, green represents the FN, and red represents the FP.

where λ_1 and λ_2 represent the loss rate coefficients. By combining the BCE loss and the sparse saliency loss, the detection performance of nonsalient change regions is effectively improved, further ensuring a balanced feature representation between salient and nonsalient regions.

IV. Experimental Results and Analysis

A. Experimental Setup

1) Datasets: To verify the practical effectiveness and applicability of SPRMamba in CD tasks, we conducted experiments on five datasets. Below is an introduction to each dataset.

LEVIR-CD [73] consists of 637 pairs of RS images, each with a resolution of 1024 × 1024 pixels, containing 636 876 269 unchanged pixels and 31 066 changed pixels. It spans more than ten years, capturing rich surface building changes, such as warehouses and residential areas. We cropped the images into 256 × 256 pixel patches, with a training, validation, and test split of 7120/1024/2048, respectively.[1]

WHU-CD [74] primarily records the post-earthquake reconstruction of Christchurch, New Zealand. It, provided by Wuhan University, is specifically for CD research, with an image resolution of 0.075 m/pixel and a size of 32 507 × 15 354 pixels. Similarly, we cropped it into 256 × 256 pixel patches, with a training, validation, and test split of 6096/762/762, respectively.[2]

GZ-CD [75] is focused on Guangzhou region in China, primarily covering surface building changes, with significant seasonal impacts. The image resolution is 0.55 m/ pixel, and the image pixel range is from 1006 × 1168 to 4936 × 5224. Similarly, it was cropped into 256 × 256 pixel patches, with a training, validation, and test split of 2,834/400/325, respectively.[3]

Change detection dataset (CDD) [76] consists of RS images that change with the seasons. It includes various types of physical changes, such as alterations in building complexes or small structures like cars, as well as seasonal variations in the natural environment, such as changes from individual trees to forested areas. Similarly, the dataset is cropped into 16 000 image pairs of size 256 × 256, with training, validation, and testing sets divided as follows: 10 000/3000/3000, respectively.[4]

Sun yat-sen university-change detection dataset (SYSU-CD) [77] contains 20 000 pairs of 0.5-m/pixel resolution images from Hong Kong, spanning from 2007 to 2014. The dataset primarily features changes in building complexes, vegetation, and road expansion. It consists of a total of 20 000 image pairs of size 256 × 256, with training, validation, and testing sets divided as follows: 12 000/4000/4000, respectively.[5]

2) Implementation Environment: The model was developed, and experiments were conducted using the PyTorch framework (version 2.1.1). The implementation utilized Python 3.8. All models were trained and evaluated on an NVIDIA RTX 4090 GPU with 24 GB, running on a system with CUDA version 12.5. Key libraries included NumPy (version 1.26.0).

[1]https://justchenhao.github.io/LEVIR

[2]https://study.rsgis.whu.edu.cn/pages/download/building_dataset.html

[3]https://github.com/daifeng2016/Change-Detection-Dataset-for-High-Resolution-Satellite-Imagery

[4]https://aistudio.baidu.com/aistudio/datasetdetail/89523

[5]https://github.com/liumency/SYSU-CD

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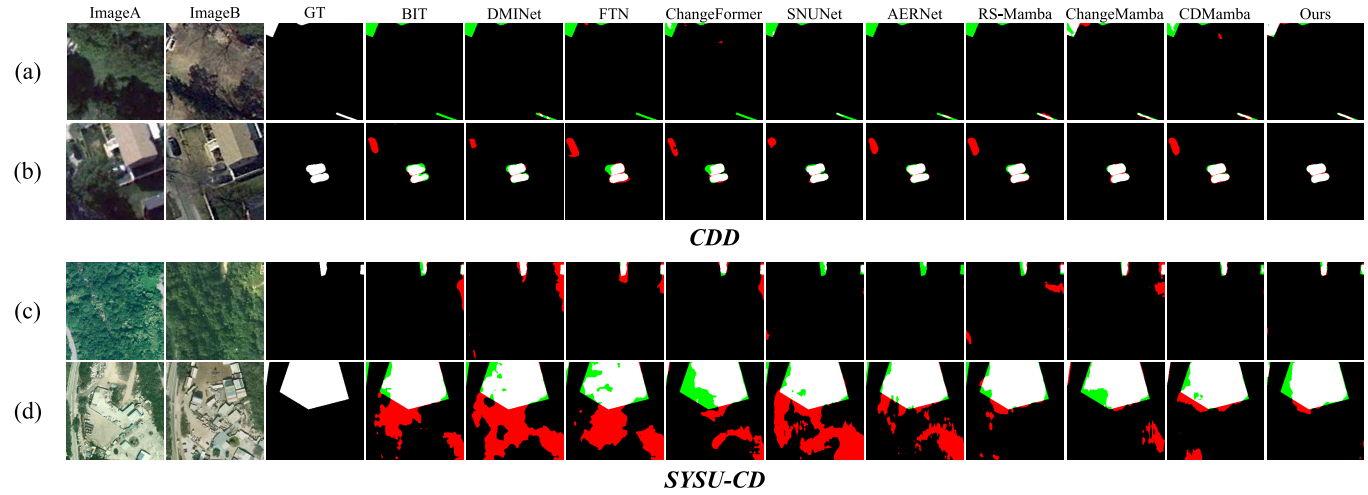


Fig. 9. Visualization results of subtle change and background-similar change areas selected from different datasets. (a) and (b) Results of different methods tested on CDD and (c) and (d) results tested on SYSU-CD, where white represents the TP, black represents the TN, green represents the FN, and red represents the FP.

TABLE II

EXPERIMENTAL RESULTS OF SPRMAMBA AND OTHER MODELS ON EACH DATASET. RED INDICATES THE HIGHEST VALUE FOR THE SELECTED METRICS, WHILE BLUE INDICATES THE SECOND-HIGHEST VALUE

Model	Publications	CDD F1 / Pre / Rec / OA / IoU (↑)	SYSU-CD F1 / Pre / Rec / OA / IoU (↑)	Params(M)↓	Flops(G)↓
BIT [25]	TGRS·2022	92.58 / 92.03 / 93.14 / 98.33 / 86.19	74.32 / 81.75 / 68.12 / 88.90 / 59.13	3.55	10.60
ChangeFormer [24]	IGARSS·2022	90.72 / 89.94 / 91.53 / 96.67 / 83.02	74.41 / 77.99 / 71.14 / 88.46 / 59.24	41.03	202.87
FTN [70]	ACCV·2022	91.02 / 91.20 / 90.84 / 97.89 / 83.52	79.41 / 82.74 / 76.33 / 90.66 / 65.85	168.53	45.00
DMINet [54]	TGRS·2023	92.91 / 91.84 / 94.00 / 98.99 / 86.76	80.06 / 84.93 / 75.72 / 91.11 / 66.75	6.24	14.55
SNUNet [53]	GRSL·2021	93.62 / 92.74 / 94.52 / 98.58 / 88.01	73.14 / 74.09 / 72.21 / 87.49 / 57.66	12.03	54.88
AERNet [71]	TGRS·2023	93.94 / 92.29 / 95.66 / 98.45 / 88.58	80.26 / 82.50 / 78.13 / 90.93 / 67.02	25.36	12.82
RS-Mamba [30]	TGRS·2024	95.21 / 93.34 / 97.17 / 98.85 / 90.87	80.43 / 85.76 / 75.73 / 91.31 / 67.27	27.90	15.71
ChangeMamba [31]	TGRS·2024	95.69 / 94.02 / 97.41 / 98.89 / 91.73	81.22 / 87.08 / 76.10 / 91.49 / 68.38	17.13	45.74
CDMamba [34]	TGRS·2025	94.57 / 92.73 / 96.49 / 98.77 / 89.70	80.58 / 86.53 / 75.40 / 91.22 / 67.48	11.90	29.64
Ours(SPRMamba)	—	96.39 / 94.40 / 98.47 / 99.13 / 93.03	82.02 / 85.55 / 78.78 / 91.86 / 69.53	23.41	11.59

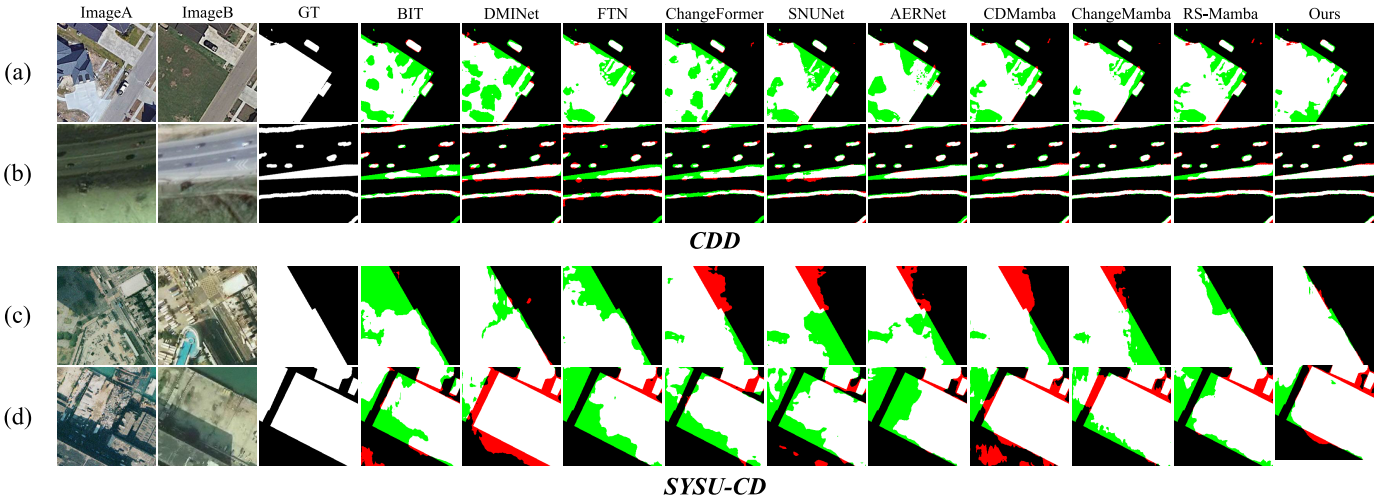


Fig. 10. Visualization results of large-scale change and dense region change areas selected from different datasets. (a) and (b) Results of different methods tested on CDD and (c) and (d) results tested on SYSU-CD, where white represents the TP, black represents the TN, green represents the FN, and red represents the FP.

During experimentation, we employed the AdamW optimizer with an initial learning rate of $5e^{-4}$ for all datasets. A linear decay learning rate scheduler was used to dynamically adjust the learning rate. Model performance was assessed at each

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TABLE III
COMPARISONS OF EFFICIENCY. PARAMETERS, FLOPS, AND TIME FOR EACH METHOD ON WHU-CD

Model	Params(M)↓	Flops(G)↓	Times(s)↓
BIT [25]	3.55	10.60	147.5
ChangeFormer [24]	41.03	202.87	373.4
FTN [70]	168.53	45.00	421.7
DMINet [54]	6.24	14.55	77.2
SNUNet [53]	12.03	54.88	405.3
AERNet [71]	25.36	12.82	265.8
RS-Mamba [30]	27.90	15.71	341.2
ChangeMamba [31]	17.13	45.74	233.6
CDMamba [34]	11.90	29.64	273.5
Ours(SPRMamba)	23.41	11.59	203.8

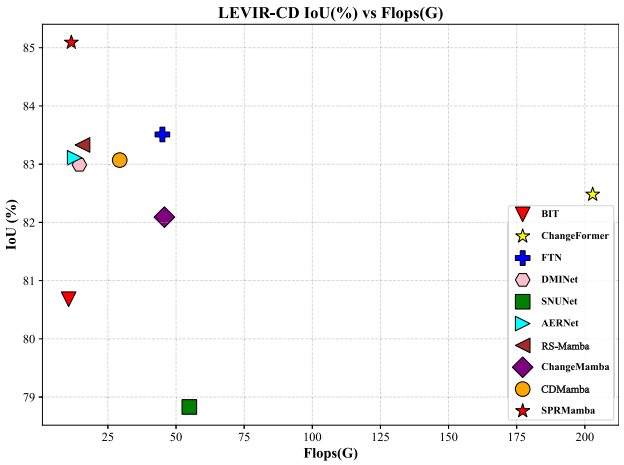


Fig. 11. Comparison of Flops and IoU across all models on LEVIR-CD.

training epoch, and the model achieving the best performance on the validation set was selected for final testing. To further validate the model's generalization and performance across different test datasets, various data augmentation techniques were implemented. For a fair comparison, the code and model weights of all comparative methods were sourced from their official open-source links provided by the authors. For methods where pretrained weights were unavailable, we reimplemented them using the same optimizer, learning rate, and scheduler settings as our proposed method to ensure experimental fairness.

3) Evaluation Metrics: We assessed model performance using five standard metrics: $F1$-score ($F1$), precision (Pre), recall (Rec), overall accuracy (OA), and intersection over union (IoU). These are based on true positives (TPs: correctly identified changes), FPs (background misclassified as change), true negatives (TNs: correctly identified background), and false negatives (FNs: missed changes). Pre measures the accuracy of positive predictions (i.e., what proportion of predicted changes are correct). Rec indicates the proportion of actual changes correctly identified by the model. $F1$ balances precision and recall, providing a single measure of overall detection accuracy. IoU quantifies the overlap between predicted and ground truth change regions. OA reflects the proportion of all pixels correctly classified. Both $F1$ and IoU are considered key performance indicators in CD tasks.

B. Comparative Experiment

1) Comparison With SOTA Methods: In this section, we compare SPRMamba with other advanced CD methods to verify its performance in CD tasks. Methods combining CNN and attention mechanisms and methods based on Transformer dominate the field. CNN and attention mechanism-based methods include DMINet [54], SNUNet [53], and AERNet [71]; Transformer-based methods include BIT [25], ChangeFormer [24], and FTN [70]; and SSM-based methods include CDMamba [34], ChangeMamba [31], and RS-Mamba [30].

Tables I and II present the performance of SPRMamba compared to other SOTA methods across different datasets. In the CD task, $F1$ and IoU metrics typically represent overall detection performance. SPRMamba improves $F1$-scores by 0.94%, 1.09%, 2.12%, 0.70%, and 0.80% compared to the second-best results on each dataset. SPRMamba also outperforms the second-best IoU scores by 1.58%, 1.95%, 3.31%, 1.30%, and 1.15%. Additionally, SPRMamba demonstrates strong competitiveness across other metrics. However, it shows second-best performance in certain specific metrics due to task-driven tradeoffs in the design of the method. Specifically, for the Rec metric, SPRMamba only lags behind AERNet on LEVIR-CD, but the Pre metric increases by 4.2%, achieved through dynamic weight decay in the SPR, which actively suppresses false change areas and achieves a better Rec–Pre balance.

For the OA metric, SPRMamba is slightly lower than CDMamba on WHU-CD. OA is sensitive to minor FP changes, and SPRMamba's window-based SSM prioritizes the continuity of local features, which may reduce the global statistical consistency and thus impact the OA score. Additionally, the sparse saliency loss can regularize the model to prevent overdetection, improving the Pre performance, but it slightly increases the risk of under-detection, leading to a decrease in Rec performance.

To visually present the test results, we provide partial data visualizations from five datasets. To comprehensively assess the model, we selected four types of images from each dataset: subtle changes, background-similar changes, large-scale changes, and dense region changes, as shown in Figs. 7–10.

Figs. 7 and 9 display the visual comparison results for subtle and background-similar change scenarios. Subtle change regions are often obscured or ignored due to weak feature signals. Specifically, in Figs. 5(a) and (e) and 7(a), the true change regions are small, and while some models can detect portions of these areas, they fail to preserve the integrity of the features. In Figs. 5(c) and 7(a), the true change features are not sufficiently distinguished from false changes. In background-similar change scenarios, environmental factors often lead to significant FPs and missed detections. For example, in Fig. 5(b) and (f), lighting interference causes noise to be misclassified as change regions. In Figs. 5(d) and 7(d), the influence of background color makes it difficult for other models to handle large homogeneous backgrounds, either

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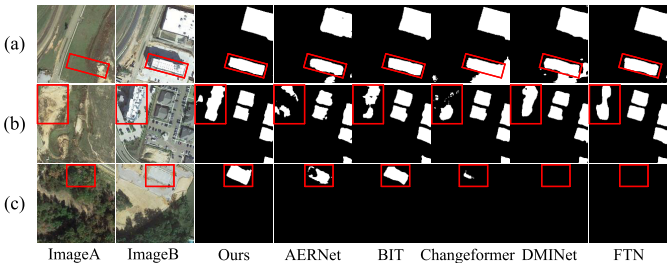


Fig. 12. Visualization of comparative examples of selected models in real-world change scenarios. (a) and (b) Represent regions disturbed by background noise, and (c) represent regions with similar backgrounds.

TABLE IV

PERFORMANCE ANALYSIS OF ABLATION EXPERIMENTS FOR SW-MAMBA AND SPR

Model	LEVIR-CD		WHU-CD		GZ-CD	
	F1↑	IoU↑	F1↑	IoU↑	F1↑	IoU↑
Base	90.81	83.17	91.97	85.13	83.11	71.10
W/O SW-Mamba	91.71	84.69	93.46	87.73	86.43	76.11
W/O SPR	91.54	84.40	93.76	88.25	85.57	74.78
SPRMamba	**91.95**	**85.09**	**94.85**	**90.21**	**87.70**	**78.10**

ignoring subtle color differences or misclassifying them as true changes, leading to false change perception biases. In contrast, SPRMamba accurately identifies subtle change regions and effectively distinguishes true changes from background noise, successfully suppressing false detections while ensuring the integrity of detected change regions.

Figs. 8 and 10 show the visual comparisons for large-scale and dense change regions. In datasets characterized by building changes, most methods can capture large-scale surface changes but struggle with boundary handling. For example, in LEVIR-CD, methods like ChangeFormer and SNUNet often cause boundary blurring or discontinuity at the edges, thus compromising the detailed perception of change regions. In dense change regions, such as the dense building scenes in WHU-CD and localized dense change scenes in GZ-CD, other methods often face issues with insufficient feature resolution, resulting in incomplete boundaries or unclear separation between change regions. CDD and SYSU-CD feature a wide variety of changes, further testing the model's boundary perception capabilities. In contrast, SPRMamba not only accurately captures large-scale change regions but also precisely refines the boundaries of dense change regions.

In summary, SPRMamba excels across all datasets, particularly in detecting nonsalient change regions and refining the boundaries of salient change areas. Through the collaborative modeling of SW-Mamba and the dynamic weighting balance mechanism of SPR, SPRMamba achieves a balanced modeling of both salient and nonsalient features, effectively reducing FN and FP. The experimental results demonstrate that SPRMamba maintains high detection accuracy and robustness across different scales, densities, and complex background conditions, proving its wide applicability and advanced performance in CD tasks.

2) Efficiency Comparison: Table III presents the Params and Flops of all models, along with their per-epoch runtime (times) on WHU-CD when the batch size is 8. Additionally, Fig. 11 compares the Flops and IoU of all models on LEVIR-CD.

From Table III, it is evident that there are significant differences in efficiency across the various backbone networks. For instance, models such as BIT, DMINet, and SNUNet, which utilize CNNs as feature extractors, have relatively low Params and Flops, but their performance lags behind more complex models. In contrast, models with Transformer backbones (e.g., ChangeFormer and FTN) or utilizing the Mamba architecture (e.g., ChangeMamba and RS-Mamba) achieve notable improvements in detection accuracy through global modeling mechanisms like attention or raster scanning. However, this comes at the cost of higher Flops or longer inference times.

Notably, SPRMamba achieves an outstanding balance between efficiency and performance. This efficiency is largely attributed to the window-based operations and cross-window interaction mechanisms in the SW-Mamba architecture, which reduce redundant computations while ensuring precise modeling of both salient and nonsalient change regions.

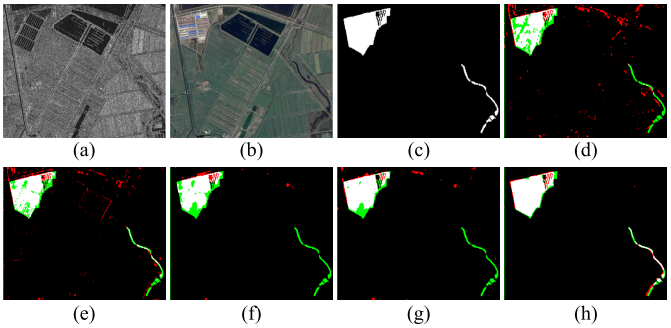


Fig. 13. Visualization results of Shuguang dataset. (a) and (b) Bi-temporal images, (c) GT, (d) output of ChangeFormer, (e) output of SNUNet, (f) output of AERNet, (g) output of ChangeMamba, and (h) output of SPRMamba.

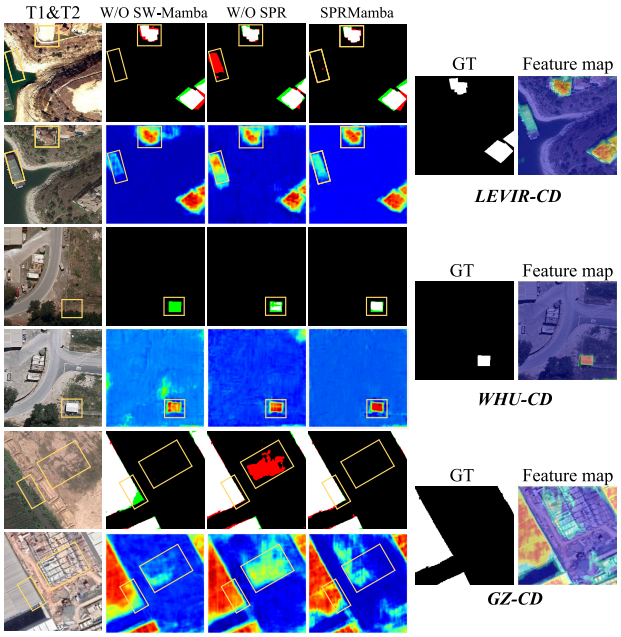


Fig. 14. Ablation Visualization Comparison on different datasets.

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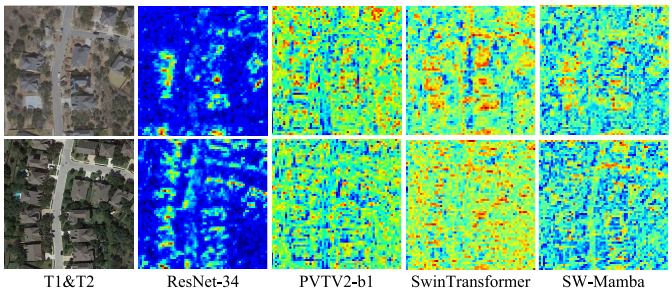


Fig. 15. Visualization of feature extraction in the first stage under different backbones.

TABLE V

PARAMETERS AND FLOPS FOR EACH BACKBONE ON THE DIFFERENT DATASETS ALONG WITH THE $F1$ AND IOU VALUES

backbone	Param(M)↓	Flops(G)↓	LEVIR-CD		WHU-CD		GZ-CD	
			F1↑	IoU↑	F1↑	IoU↑	F1↑	IoU↑
ResNet34	26.10	12.42	91.38	84.13	92.26	85.64	84.22	72.73
PVTV2_b1	22.32	10.78	91.55	84.41	93.04	86.99	85.14	74.13
Swin-Transformer	41.96	17.26	91.71	84.69	93.46	87.73	86.43	76.11
SW-Mamba	23.41	11.59	**91.95**	**85.09**	**94.85**	**90.21**	**87.70**	**78.10**

Overall, SPRMamba significantly optimizes complexity while maintaining detection accuracy, making it suitable for large-scale CD tasks in RS applications.

3) Comparison of Real-World Change Scenarios: Data variability remains a fundamental challenge in CD tasks, making the construction of highly generalized network architectures crucial for adapting to complex change scenarios. We validate the detection capability of SPRMamba in real-world settings using Google Earth imagery from Oxford, MS, USA (2012–2019), as illustrated in Fig. 12.

Considering the dynamic complexity of real-world land cover and building structures, we directly evaluate models with optimal weights trained on the LEVIR-CD dataset. As a benchmark CD dataset, LEVIR-CD effectively represents typical change types encountered in practical scenarios. Without fine-tuning, all methods are directly applied to real CD tasks [26]. While all models successfully capture prominent building changes in Fig. 12(a), detection difficulty escalates significantly in background-interference regions. For the subtle changes and single-object variations demonstrated in Fig. 12(b) and (c), existing methods suffer from edge ambiguity and background-like noise.

4) Comparison of Shuguang Dataset: Through multiscale feature fusion mechanisms, SPRMamba enhances weak change responses while effectively suppressing false alarms, demonstrating superior edge preservation and scenario adaptability.

Shuguang dataset [78], a heterogeneous dataset involving multisensor scenarios, poses significant challenges for CD due to variations in imaging conditions and spectral features, as shown in the visualization results in Fig. 13. Existing methods exhibit notable differences in handling the dynamic changes of two typical land cover types, buildings, and rivers. Buildings, with their regular geometric shapes and stable spectral responses, typically yield relatively complete detection results.[6]

[6]https://github.com/yulisun/INLPG/tree/master/datasets

TABLE VI

COMPARISON OF $F1$ AND IOU VALUES BETWEEN G-SSM AND L-SSM IN SW-MAMBA ON EACH DATASET

Module	G-SSM	L-SSM	LEVIR-CD		WHU-CD		GZ-CD	
			F1↑	IoU↑	F1↑	IoU↑	F1↑	IoU↑
SW-Mamba	✓	×	91.61	84.51	93.34	87.50	85.43	74.56
SW-Mamba	×	✓	91.70	84.67	93.78	88.29	86.35	75.98
SW-Mamba	✓	✓	**91.95**	**85.09**	**94.85**	**90.21**	**87.70**	**78.10**

TABLE VII

COMPARISON OF THE EFFECTS OF WINDOWING OPERATIONS, MULTILEVEL SHIFTING, AND WINDOW SIZES ON THE GZ-CD DATASET

Windows	H-Shift	Size	GZ-CD	
			F1↑	IoU↑
×	-	4	83.35	71.46
✓	✓	4	85.22	74.24
✓	×	8	86.09	75.58
✓	✓	8	**87.70**	**78.10**
×	-	16	84.05	72.49
✓	✓	16	86.16	75.68

In contrast, river regions are affected by factors such as light reflection and background similarity, leading to fuzzy boundaries and low contrast, which make detection more challenging.

Specifically, ChangeFormer and SNUNet are interfered with by road texture similarities, leading to strip-like misdetections in riverbank areas. While AERNet and ChangeMamba effectively suppress discrete noise, they show insufficient sensitivity to fragmented changes caused by gradual water body expansion.

SPRMamba innovatively integrates SW-Mamba and SPR, leveraging a multiscale harmonization mechanism to simultaneously enhance the representation of large-scale building structure changes and fine-scale dynamic river transformations. It significantly improves the capture of fine-grained changes in low-contrast scenes while maintaining the integrity of salient change regions.

C. Ablation Study

To demonstrate the targeted performance improvements brought by the innovative modules in SPRMamba, we conducted various ablation experiments and visual demonstrations.

1) Effectiveness of Model Innovations: Through ablation experiments and visual analysis, we validated the effectiveness of the SW-Mamba and SPR modules in the CD task, as shown in Table IV and Fig. 14. Specifically, SPRMamba achieved $F1$-scores and IoU values of 91.95%/85.09%, 94.85%/90.21%, and 87.93%/78.46% on different datasets, demonstrating its outstanding performance in CD tasks.

In Fig. 14, when SW-Mamba is removed (W/O SW-Mamba) and Swin-Transformer [64] is used as the backbone, both models adopt a hierarchical window partitioning strategy, thus eliminating the interference from window parameter differences. However, the model loses its ability to effectively model weak features, as SW-Mamba enhances the

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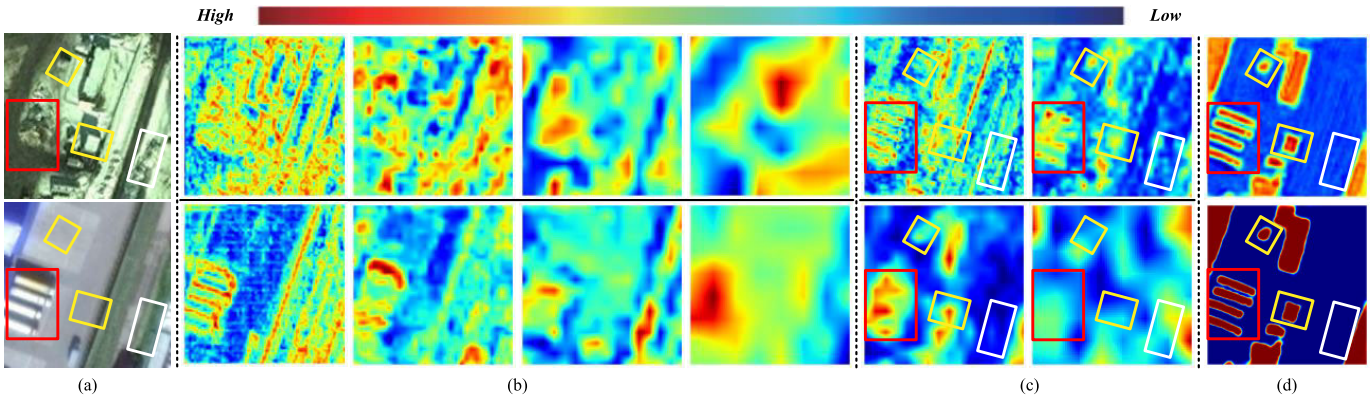


Fig. 16. Visualization of feature maps from different stages of SPRMamba. The boxes outlined in red, yellow, and white represent salient changes, nonsalient changes, and pseudo changes, respectively. (a) Bi-temporal input images, (b) feature outputs from four stages obtained through SW-Mamba, (c) feature outputs from four stages processed by SPR, and (d) final output feature map along with the output after applying the sigmoid activation function.

TABLE VIII
ABLATION ANALYSIS OF DIFFERENT RECURSION LEVEL SETTINGS IN SPRMAMBA

Stage 1	Stage 2	Stage 3	Stage 4	Params(M)↓	Tt(s)↓	WHU-CD	
						F1↑	IoU↑
$L_1 = 2$	$L_2 = 2$	$L_3 = 2$	$L_4 = 2$	20.76	177.4	92.35	85.78
$L_1 = 2$	$L_2 = 2$	$L_3 = 4$	$L_4 = 2$	23.41	203.8	94.85	90.21
$L_1 = 2$	$L_2 = 2$	$L_3 = 8$	$L_4 = 2$	28.71	365.5	93.13	87.14
$L_1 = 2$	$L_2 = 2$	$L_3 = 16$	$L_4 = 2$	39.31	480.3	**94.92**	**90.34**
$L_1 = 2$	$L_2 = 2$	$L_3 = 18$	$L_4 = 2$	41.96	531.6	94.81	90.13

TABLE IX
COMPARISON OF DIFFERENT RECURSION LEVEL SETTINGS IN SPR

Stage 1	Stage 2	Stage 3	Stage 4	LEVIR-CD	
				F1↑	IoU↑
×	×	×	×	91.54	84.40
✓	×	×	×	91.66	84.60
✓	✓	×	×	91.72	84.71
✓	✓	✓	×	91.78	84.79
✓	✓	✓	✓	**91.95**	**85.09**

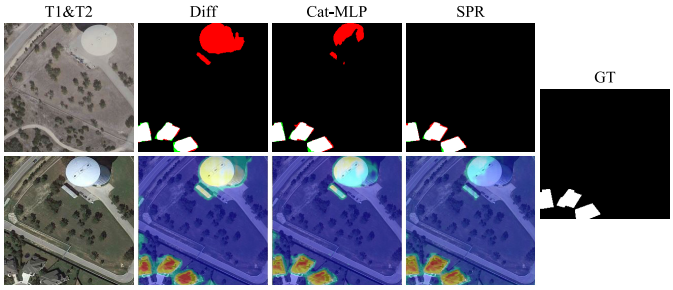


Fig. 17. Visual comparison of different bi-temporal feature fusion methods.

TABLE X
COMPARISON OF DIFFERENT FUSION METHODS FOR BI-TEMPORAL IMAGES

Method	Param(M)↓	FLOPs(G)↓	LEVIR-CD		WHU-CD		GZ-CD	
			F1↑	IoU↑	F1↑	IoU↑	F1↑	IoU↑
Diff	14.14	8.97	91.22	83.85	91.86	84.95	83.55	71.75
Cat-MLP	20.89	9.77	91.72	84.71	93.78	88.29	86.89	76.82
SPR	23.41	11.59	**91.95**	**85.09**	**94.85**	**90.21**	**87.70**	**78.10**

representation of weak features through window compression and interaction mechanisms, significantly reducing missed detections in nonsalient change regions. When SPR is removed (W/O SPR), the model is influenced by irrelevant information, as SPR effectively separates salient and nonsalient features through a dynamic saliency balancing strategy, suppressing background noise interference and refining edge detection.

With the complete configuration, the SPRMamba model achieves the best performance on the LEVIR-CD, WHU-CD, and GZ-CD datasets. It accurately captures both salient and nonsalient changes in complex scenarios, demonstrating exceptional robustness and accuracy.

2) Effectiveness of SW-Mamba: In CD tasks, the feature extraction capabilities of different backbones are significantly different. To fairly evaluate the performance, we choose the backbone with similar magnitude of SW-Mamba parameters for comparative analysis, as shown in Fig. 15.

Experiments show that the feature distribution of ResNet34 is relatively sparse, and its response is mainly concentrated in the salient change region, which makes it difficult to capture weak features. Although PVTV2-b1 improves the feature density, it introduces irrelevant noise and redundant information, which leads to the confusion of salient/nonsalient region features.

The Swin-Transformer enhances the response of the target region through the sliding window mechanism, but the response is intense due to the attention concentration effect, which increases the difficulty of extracting the salient and nonsalient features.

In contrast, SW-Mamba shows unique advantages: 1) the linear interaction mechanism based on SSM can suppress background noise while retaining weak features; 2) overlay scanning with dynamic window shift mechanism to enhance the detection ability of tilted edges and irregular changes; and 3) the dual-branch architecture [global state space model (G-SSM) + local state space model (L-SSM)] realizes the collaborative modeling of global semantics and local details, while the single-branch design of Swin-Transformer is difficult to balance this contradiction.

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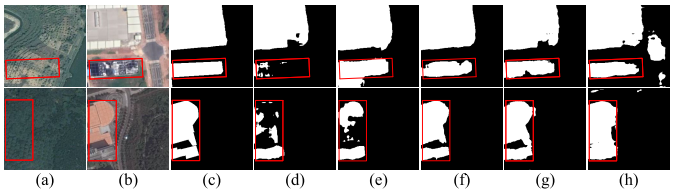


Fig. 18. Impact of different loss function parameters. (a) and (b) Bi-temporal images; (c) ground truth; (d) parameter configuration as 0.8, 0.5, and 0.5; (e) parameter configuration as 0.8, 0.6, and 0.4; (f) parameter configuration as 0.5, 0.7, and 0.3; (g) parameter configuration as 0.5, 0.6, and 0.4; and (h) parameter configuration as 1, 0.7, and 0.3.

TABLE XI
INFLUENCE OF λ_1 AND λ_2 ON $F1$ AND IOU METRICS ACROSS DIFFERENT DATASETS

λ_1	λ_2	LEVIR-CD		WHU-CD		GZ-CD	
		F1↑	IoU↑	F1↑	IoU↑	F1↑	IoU↑
1	1	91.69	84.65	92.47	86.00	85.24	74.28
0.5	1	91.56	84.43	91.94	85.09	84.28	72.83
1	0.8	91.73	84.72	93.75	88.24	85.57	74.78
1	0.5	**91.95**	**85.09**	**94.85**	**90.21**	**87.70**	**78.10**
0.5	0.5	91.81	84.86	94.19	89.02	86.97	76.95
1	0.2	91.70	84.67	93.59	87.95	85.88	75.26
1	0	91.60	84.50	92.22	85.56	84.46	73.10

As shown in Table V, SW-Mamba still leads significantly with 94.85% $F1$ under the premise that the computational complexity is reduced by 44% compared with that of Swin-Transformer, which confirms its double breakthrough in feature expression balance and computational efficiency.

SW-Mamba consists of a pooled cross-window global modeling branch (G-SSM) and a windowed local modeling branch (L-SSM). Table VI demonstrates the contributions of G-SSM and L-SSM. When using only G-SSM, the model exhibits strong global modeling capabilities for salient change regions but struggles to capture fine details in nonsalient change regions, leading to missed detections and blurred edges.

In contrast, using only L-SSM enables excellent local detail modeling but lacks the integration of global information, resulting in incomplete perception of large-scale changes. The combined G-SSM and L-SSM in SW-Mamba effectively achieves balanced modeling of global information and local details, showcasing its superior performance.

Table VII illustrates the effects of windowing operations, the multilevel dynamic shifting mechanism, and window size in SW-Mamba. The results demonstrate that both the windowing operation and the shifting mechanism are essential.

Additionally, the choice of window size is critical; a moderate window size strikes a balance between global modeling and fine-grained feature representation, delivering optimal performance across various scenarios.

Table VIII reveals the critical effect of the SW-VSS block depth configuration. When the number of SSM layers in Stage 3 ranges from $L_3 = 2$ to $L_3 = 16$, $F1$ on WHU-CD improves by 2.57%, indicating that deep state-space modeling effectively captures long-range cross-window dependencies.

TABLE XII
INFLUENCE OF λ_2, μ, AND θ ON $F1$ AND IOU METRICS ACROSS DIFFERENT DATASETS

λ_2	μ	θ	LEVIR-CD		WHU-CD		GZ-CD	
			F1↑	IoU↑	F1↑	IoU↑	F1↑	IoU↑
0.8	0.5	0.5	91.63	84.56	92.88	86.70	84.46	73.09
0.8	0.6	0.4	91.73	84.72	93.75	88.24	85.57	74.78
0.5	0.7	0.3	**91.95**	**85.09**	**94.85**	**90.21**	**87.70**	**78.10**
0.5	0.6	0.4	91.88	84.98	93.45	87.70	86.63	76.41
1	0.7	0.3	91.69	84.65	92.47	86.00	85.24	74.28

3) Effectiveness of SPR: However, when $L_3 = 18$, the performance improvement over the initial configuration becomes limited, and the computational overhead reaches its maximum. Although $L_3 = 16$ achieves peak accuracy ($F1 = 94.92\%$), its computational cost increases by 160% compared to $L_3 = 4$ (203.8 s). Therefore, for practical deployment, $L_3 = 4$ ($F1 = 94.85\%$) can be selected as the optimal efficiency–accuracy tradeoff based on resource constraints.

Fig. 16 illustrates the visualization of the SPRMamba workflow. The bi-temporal input images are first processed by the SW-Mamba hierarchical encoder to extract multiscale features. Then, each stage is decoded and integrated through SPR, and finally, the pixel-level high-resolution change map (Final Map) is gradually aggregated through upsampling. During the decoding stage, SPR effectively preserves weak features, aggregating shallow details and deep semantics to balance the representation of real change regions.

Table IX illustrates the impact of SPR on model performance at different stages. Stages 1–4 correspond to the feature fusion processes at dimensions 64, 128, 256, and 512, respectively. Specifically, introducing SPR only at the first stage yields significantly inferior results compared to its implementation in two stages or full integration. As the number of SPRs increases, the model achieves better multiscale dynamic feature fusion, enhancing its ability to perceive both salient and nonsalient change regions. This improvement leads to greater accuracy and robustness in the detection results.

Table X and Fig. 17 present the impact of different bi-temporal feature fusion methods on model performance. Here, Diff represents the direct difference processing of multilevel bi-temporal images, while Cat-MLP denotes concatenating multilevel bi-temporal features followed by dimensionality reduction using MLP. Specifically, Diff shows the poorest performance, as the direct difference operation tends to introduce irrelevant region interference, limiting its effectiveness in handling complex scenarios. In comparison, the MLP-based dimensionality reduction method performs better in capturing the nonlinear relationships between features.

In contrast, our SPR effectively suppresses noisy regions caused by environmental interference, enabling the precise identification of true change regions.

4) Effectiveness of Loss Function Parameters: The impact of different loss function parameter combinations on model performance is shown in Tables XI and XII. For SPRMamba, the loss calculation integrates both the final change feature map and the intermediate change feature maps at each stage.

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In Table XI, the value of λ_2 reflects the sparse saliency loss calculation for the intermediate feature maps.

Increasing λ_2 appropriately leads to improvements in $F1$ and IoU across various datasets, indicating that moderate sparse saliency constraints and appropriate weight distribution for salient regions contribute to balancing feature representation. In contrast, configurations like $\lambda_1 = 0.5$ and $\lambda_2 = 1$ show a decline in performance, suggesting that overly strong sparsity inhibits the transmission of key features. In Table XII, the value of μ emphasizes the sensitive capture of salient features, while the value of θ ensures a smooth transition for weak change regions. The optimal performance ($\lambda = 0.5$, $\mu = 0.7$, and $\theta = 0.3$) highlights that a reasonable parameter configuration is crucial for optimizing the balance between salient and nonsalient change regions in the model.

Fig. 18 presents the detection results under different loss function configurations. As shown in (d)–(h), when the sparse saliency loss is imbalanced, nonsalient change regions suffer from missed detections, noise, blurred edges, and noticeable false detections. As the sparse saliency loss gradually balances, the model enhances its ability to capture nonsalient change regions, reduce noise, and sharpen edge contours.

V. Discussion

The Mamba architecture, characterized by its linear complexity and selective S6 block, offers RSCD advantages, efficiently modeling global context in large-scale imagery and preserving nonsalient features often missed by CNNs or attention. However, adapting its 1-D sequential nature to 2-D spatial data is challenging; ViM's multidirectional scans may not fully capture isotropic local details, motivating our SW-Mamba to better integrate local and global information. While Mamba's S6 selection surpasses static SSMs (e.g., S4), its training stability and generalization on diverse RS sensors need further validation compared to established architectures.

Compared to attention mechanisms, Mamba offers scalable global modeling for RSCD. Its selective state recursion efficiently captures long-range dependencies, contrasting with attention's costly pairwise interactions. The challenge lies in balancing Mamba's global efficiency and subtle feature preservation with CNNs' robust local extraction and Transformers' precise but expensive interactions, guiding architectural choices for specific RSCD tasks.

VI. Conclusion

In this article, we proposed an innovative RSCD method, SPRMamba, a Mamba-based network, which effectively addressed the imbalance in feature modeling between salient and nonsalient change regions by combining the SW-Mamba backbone with the SPR and the sparse saliency loss. Specifically, the SW-Mamba backbone utilizes a window mechanism and cross-window interaction, inherent to its Mamba-inspired design, to effectively capture global semantic information and local fine-grained features, thereby enhancing the model's ability to perceive nonsalient change regions. The SPR, through a dynamic weighting strategy and differential feature enhancement, is coupled with the sparse saliency loss function during the optimization process, improving the comprehensive representation capability of both salient and nonsalient features.

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